

Gabriel Pesce

Tidiane Bengrich

Małgorzata Gierdewicz

Hallucinations in LLM

First Iteration

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What we have done? How's the work got managed into the project team?

1. Gabriel Pesce

- a. **Development of methodology:** creation of functions and classes to organize variables and results efficiently.
- b. Many issues encountered during development were successfully resolved, but the code will be refactored more in order to reduce the density of the methodology class.

2. Tidiane Bengriché

- a. **Prompts evaluation and metrics definition:** designed useful metrics to evaluate results.
- b. Large number of prompts given to the chatbots, highlighting agreements and disagreements between the two bots.

Initial insights & conclusions

Initial tests and first issues

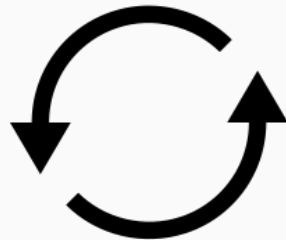
Initial issues during methodology testing:

1. **Methodology stopped during initial steps** (before the loop).
 - a. Second chatbot always returned “True” initially, even for clearly fake news. → Solved by providing more specific context in both evaluation and rewriting; models now behave more like fact-check detectors.
2. **Generated outputs were sometimes incorrect or incoherent.**
 - a. Inconsistent evaluations: a bot’s agreement with a statement was sometimes recorded incorrectly, giving the opposite result. → After bugs fixing, models’ answers are now more coherent and aligned with their evaluations.

Initial insights & conclusions

Issue during metrics test (to be addressed)

One chatbot initial disagreement would often turn into agreement during modification, only for the other to then disagree, leading to an endless cycle of changing minds and no final consensus.



Initial insights & conclusions

Metrics

4 types of results :

- If the news is false and the LLM corrects it -> **True Negative** (TN) (Good detection)
- If the news is false and the LLM accepts it as true information -> **False Positive** (FP)
- If the news is true and LLM accepts it as true information -> **True Positive** (TP)
- If the news is true and LLM rejects it as false information -> **False Negative** (FN)

Metrics :

- **Vigilance** = $TN / (TN + FP)$ (Capacity to find false news)
- **Reliability** = $TN / (TN + FN)$ (Capacity to avoid rejecting true info news)
- **F1 Score** = $2 * (Vigilance * Reliability) / (Vigilance + Reliability)$

Initial insights & conclusions

Metrics Results (over 100 iterations)

DeepSeek :

- **Vigilance** : 0.170
- **Reliability** : 0.889
- **F1 Score** : 0.286

Gemini:

- **Vigilance** : 0.180
- **Reliability** : 0.818
- **F1 Score** : 0.295

The two models handle true information well (Reliability), but they fail to detect most fake news (Vigilance).

For this first iteration, the results are not optimal. Our next goal is to test again the models and giving more detailed context, focusing more on the fake news and improve their behaviour in order to detect them.

Comparison with the baseline paper

Based on the proposed paper ([Link to the paper](#)) we can state the following pros:

1. **Metrics proposal:** while the baseline paper does not define quantitative metrics, our approach introduces a clear evaluation scheme (**TN/TP/FP/FN, Vigilance, Reliability, F1**), making correctness measurable.
2. **Prompting strategy:** while the baseline uses a straightforward single-prompt setup, our method is multi-phase (evaluation + rewriting), increasing robustness and contextual accuracy.
3. **Error-correction mechanism:** unlike the baseline, our approach actively detects inconsistencies and attempts to correct them through iterative rewriting, improving coherence and reducing misclassification.